DISHA JAIN

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Software Developer with experience building backend services, analytics platforms, and AI-powered applications. Hands-on experience with **NestJS, TypeScript, Python, ClickHouse, MongoDB, Redis, LLMs, and RAG**, with a focus on scalable backend systems and AI-assisted product features.

Education

International Institute of Information Technology (IIITH) — **Hyderabad, Telangana**
M.Tech in CSIS - CGPA: 7.52/10.00 — *July 2023 - May 2025*

Shri Vaishnav Vidyapeeth Vishwavidyalaya — **Indore, Madhya Pradesh**
B.Tech in CSE - CGPA: 8.5/10.00 — *July 2018 - May 2022*

Experience

Coschool.ai - ***Software Developer*** — **March 2025 - Present**
EdTech platform - AI-powered learning for K-12 schools (students, teachers, parents)

Role-Based Analytics Dashboards | *NestJS, ClickHouse, MongoDB, Materialized Views, AWS SES, GitHub Actions*

- Built **role-based analytics dashboards** for teachers using **NestJS, ClickHouse and MongoDB**, tracking homework completion and academic performance across classes, sections and subjects.
- Improved dashboard response times by **58%** by reducing expensive database work and using **ClickHouse materialized views** to pre-compute frequently used student activity metrics.
- Built **one-click dashboard actions** for principals and teachers to act on student insights directly, including **nudges for inactive students, login reminders, appreciation emails**, and assigning **level-up homework** to students marked **Needs Improvement**.
- Implemented **GitHub Actions CI/CD workflows** to automate testing, build validation, and deployment of NestJS-based backend services.

LLM-Based Question Generation & Validation Pipeline | *LangGraph, LLMs, Prompt Engineering, AWS S3, Milvus*

- Built an **LLM-based question generation pipeline** using **LangGraph** to generate **chapter-aligned questions** from teacher-uploaded PDFs.
- Built a **validation pipeline** to evaluate generated questions for correctness, completeness, and alignment with chapter content.
- Used an **AWS Lambda function** to fetch structured curriculum metadata, including topics and subtopics, from the knowledge tree for question generation.
- Generated **question explanations and step-by-step solutions** to help students understand each question, the underlying concept, and the complete problem-solving approach.

Student Progress Reporting System | *NestJS, MongoDB, ClickHouse, Redis*

- Developed an **automated student progress reporting system for 90K+ students**, providing parents with insights into academic performance, homework, and learning activities.
- Generated **personalized student insights** highlighting weak chapters, inactivity, overdue homework, and areas requiring attention.
- Implemented **scheduled cron jobs** for automated report generation, storing historical reports in MongoDB and using **Redis caching** for faster retrieval.

Practice & Mock Test Engine | *NestJS, TypeScript, ClickHouse, MongoDB*

- Developed an **end-to-end Practice and Mock Test engine** supporting pause/resume, cross-device continuation, saved answers, time limits, and inactivity handling.
- Implemented **single active test session handling** to prevent duplicate attempts and conflicting answers across multiple devices or tabs.
- Built **post-test analytics and reports** covering score, accuracy, time spent per question, revisits, and answer changes to help students identify areas for improvement.

Personal Projects

P2P File Sharing System | *C++, Socket Programming, Multi-threading*

- Built a **group-based peer-to-peer file sharing system** using socket programming, user authentication, **SHA-1 integrity checks**, and multi-threaded simultaneous downloads.

Scalable Media Conversion | *Python, RabbitMQ, MongoDB GridFS*

- Developed a **microservices-based video-to-audio conversion system** with JWT authentication, asynchronous processing using RabbitMQ, **MongoDB GridFS for storing video and audio files**, and email notifications.

Technical Skills

Languages:	C++, Python, TypeScript
Databases:	ClickHouse, MongoDB, MySQL, PostgreSQL, Redis
Frameworks/Tools:	NestJS, Node.js, FastAPI, Kafka, Locust, Pytest
AI:	LangGraph, LLMs, RAG, Prompt Engineering
Coursework:	Operating Systems, Computer Networks, Data Structures & Algorithms, Database Management Systems, OOPs

Achievements

- Secured **98.79 percentile** in GATE CS 2023.
- Ranked in the **top 100** in PGEE 2023.
- Qualified **BARC OCES/DGFS 2023**.
- **IIITH Dean's List**, Monsoon 2024.